



Fr. Conceicao Rodrigues College of Engineering

Full Stack Development

Topic: Virtual Lab of Python

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2024-2025

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Home page:

PyLab Home Lessons Features How It Works Code Editor Sign In [Get Started](#)

Virtual Python Laboratory

Learn Python Interactively in Your Browser

PyLab provides an interactive, hands-on environment for learning Python programming without any installations. Code, experiment, and grow your skills with real-time feedback.

[Try It Now](#) [Learn More](#)

```
python terminal
> Welcome to PyLab!
> print("Hello, World!")
Hello, World!
> |
```





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Features page:

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Features

Everything You Need to Learn Python

Powerful tools and resources to make your learning journey effective and enjoyable

- **Interactive Coding Environment**
Code directly in your browser with real-time execution and feedback
- **Step-by-Step Tutorials**
Follow guided lessons designed for beginners to advanced users
- **AI-Powered Assistance**
Get help and hints from our intelligent code assistant when you're stuck
- **Visualization Tools**
- **Challenge Projects**
- **Progress Tracking**





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Dashboard page:

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A Simple Yet Powerful Approach

Our methodology makes learning Python intuitive and enjoyable

Learn Practice Create

Learn Python Fundamentals

Start with the basics and build a solid foundation

Begin your journey with interactive lessons covering Python syntax, data types, control structures, and more. Each concept is explained with clear examples and visualizations.

Our step-by-step approach ensures you gain a deep understanding of fundamental programming concepts.

```
lesson_1.py
# Learning variables and data types
name = "Python Learner"
age = 25
is_student = True

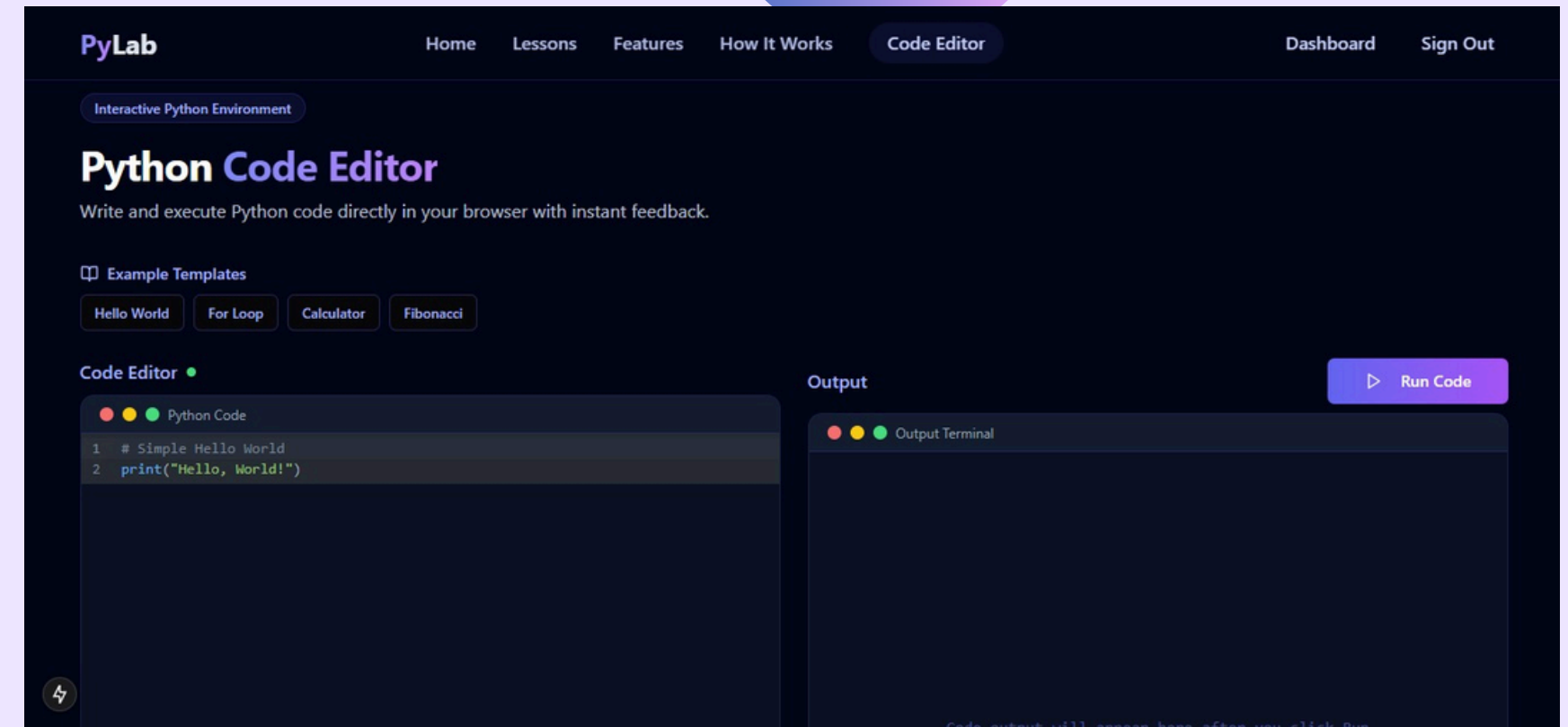
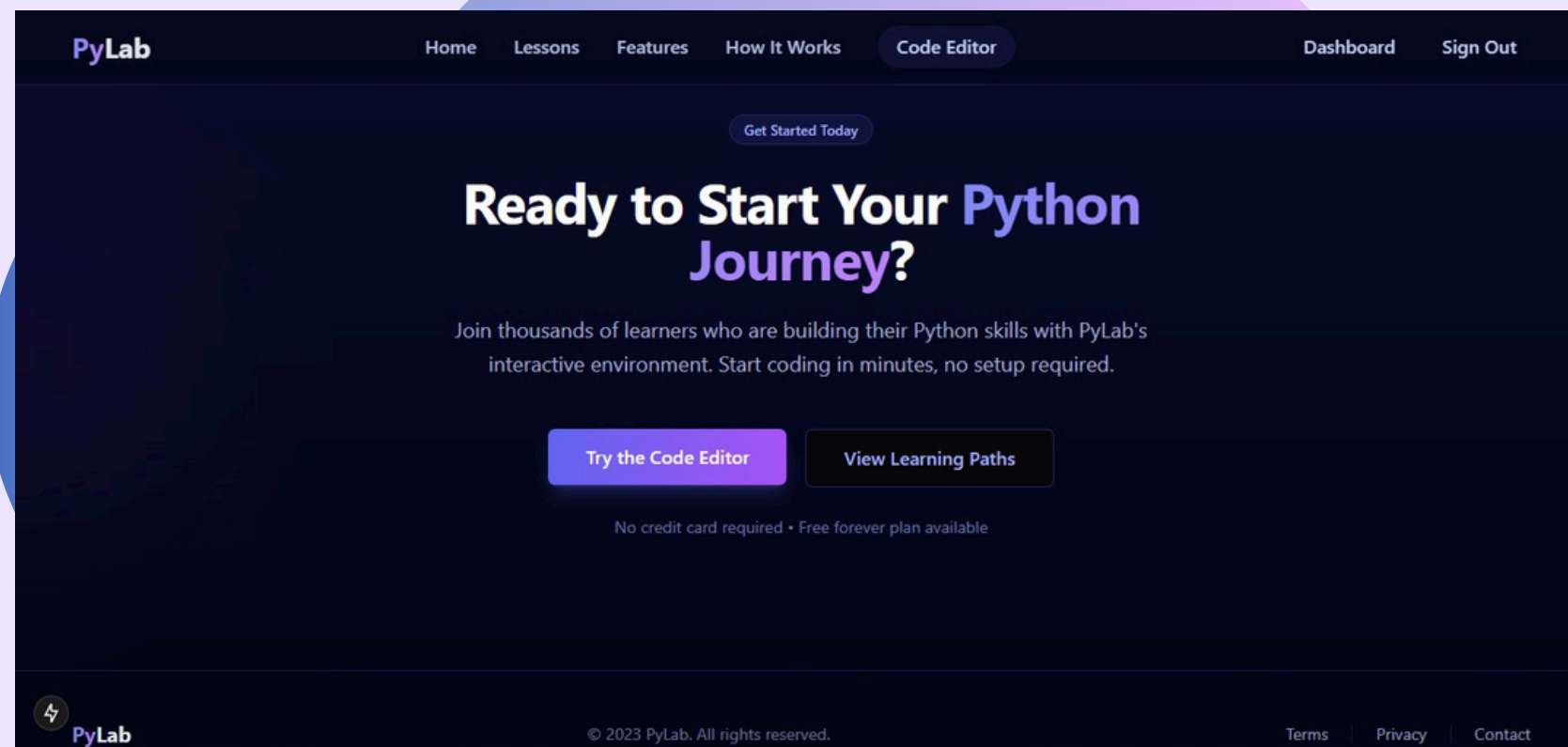
print(f"Hello, {name}! You are {age} years old.")
```





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Code Editor page:





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Admin Dashboard page:

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Admin Dashboard

Manage PyLab courses, students, and monitor progress

Metric	Value
Total Students	42
Total Lessons	18
Lesson Completions	156

Quick Actions

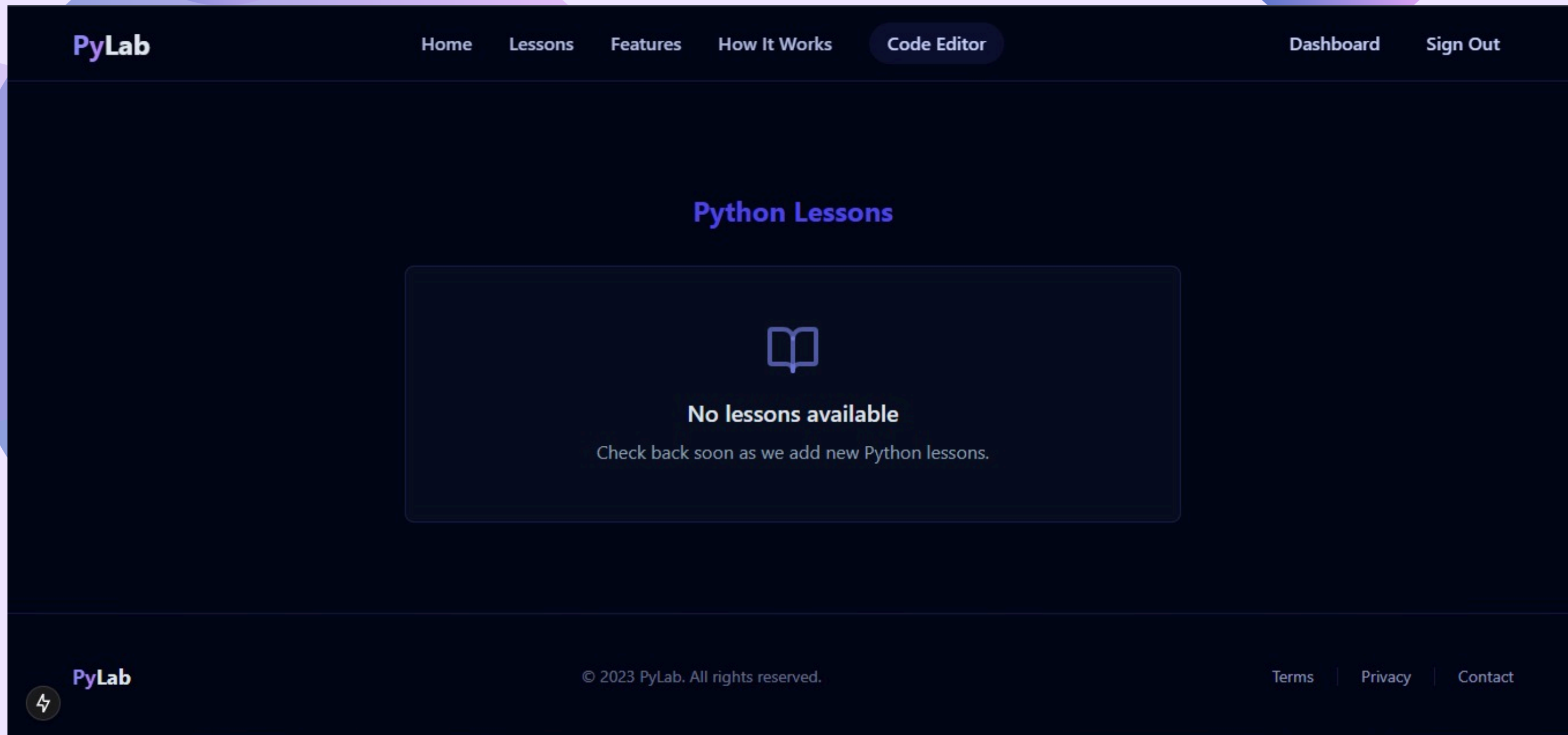
Action	Description	Button
Manage Lessons	Create, edit or delete lessons	Go to Lessons
Manage Students	View and manage student accounts	Go to Students
Test Lab Environment	Test the Python execution environment	Open Editor





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Lessons page:





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Create Lessons page:

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Create New Lesson

Add a new Python lesson to the platform

Lesson Details

Fill out the information below to create your lesson

Lesson Title
Introduction to Python Variables

Description
A brief introduction to Python variables and data types...

Lesson Content (Markdown)
Markdown supported
Introduction to Python Variables In this lesson, we'll learn about Python variables: ## What are Variables? Variables are containers for storing data values in Python...





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Create Lessons page:

PyLab

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Use Markdown for formatting. Headers (#, ##), lists (-, *), code blocks (``python), etc.

Cancel

Create Lesson

Markdown Preview Tips

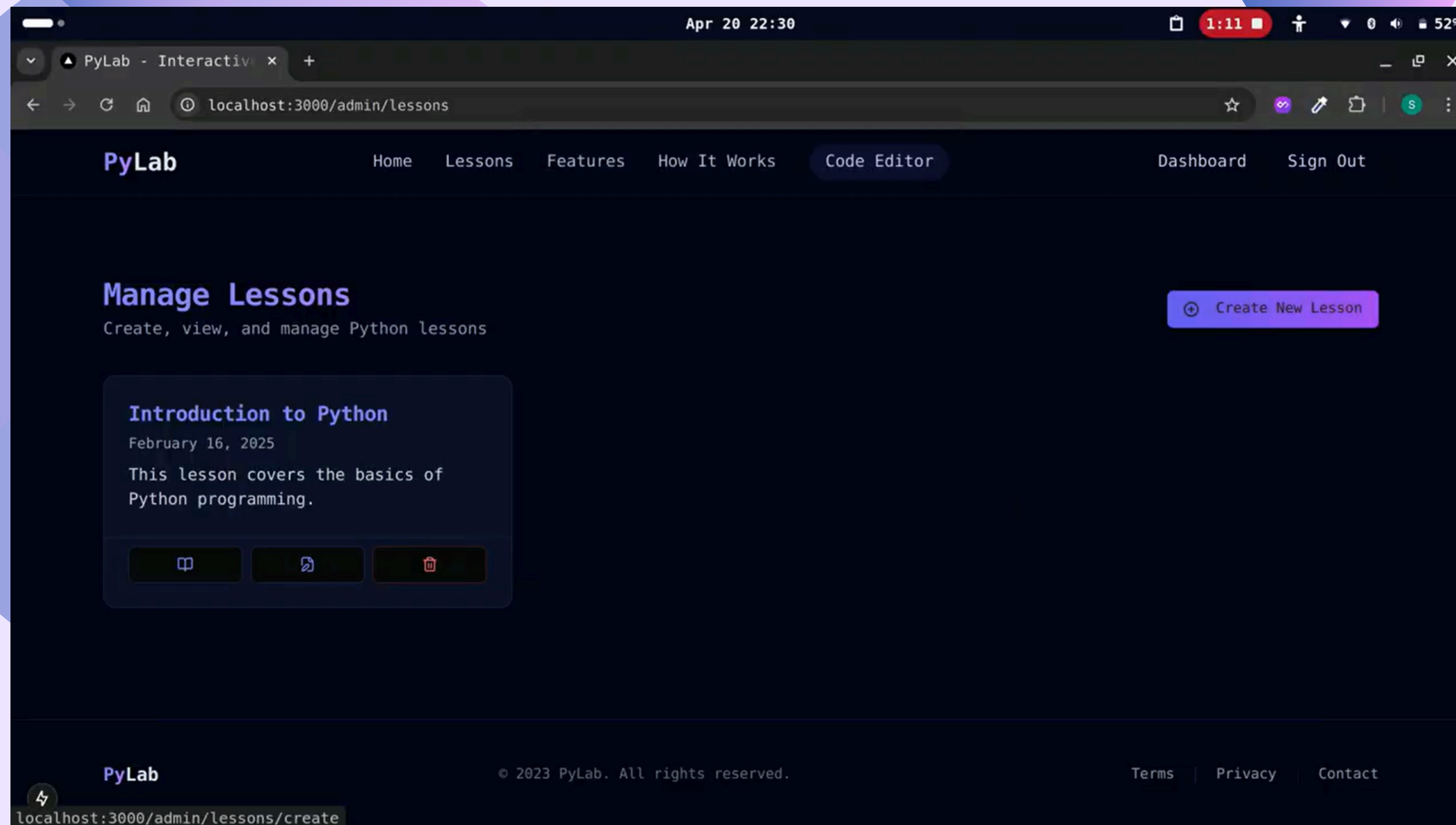
- # Heading 1 - For main titles
- ## Heading 2 - For section titles
- - Item or * Item - For bullet lists
- ```python
print("Hello World")
``` - For code blocks
- \*\*bold\*\* - For bold text





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## Manage Lessons page:





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## Create Student and Admin Account Page:

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### Create Student Account

Fill out the form below to create your account

**Name**

**Roll Number**

**Email**

**Password**

[Sign Up](#)

[Already have an account? Sign in](#)

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### Sign In as Student

Enter your credentials to access your account

**Email**

**Password**

[Sign In](#)

[Don't have an account? Sign up](#)





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## Login Student and Admin Account Page:

PyLab

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Sign In as Student

Enter your credentials to access your account

Email

Password

Sign In

Don't have an account? [Sign up](#)

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[Sign In](#) [Get Started](#)

Sign In as Admin

Enter your credentials to access your account

Email

Password

Sign In

Don't have an account? [Sign up](#)





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## Practice Segment:

The screenshot displays the PyLab website interface. At the top, there is a navigation bar with links for Home, Lessons, Features, How It Works, Code Editor, Sign In, and a Get Started button. Below the navigation bar, there are three tabs: Learn, Practice (which is highlighted), and Create. The main content area is titled "Practice with Real Exercises" and includes the subtitle "Reinforce your knowledge through practical coding challenges". The text describes the practice segment as a place to apply what's learned with hands-on exercises and projects, receive immediate feedback, and work through challenges. It also mentions that practice is key to mastering programming and that challenges range from simple to complex. On the right side, there is a code editor window titled "exercise\_2.py" containing Python code for a function to find the maximum number in a list. The code is as follows:

```
exercise_2.py
Exercise: Create a function to find the max numl
def find_max(numbers):
 # Your code here
 pass

Test your function
result = find_max([5, 12, 9, 42, 3])
print(f"Max number: {result}")
```







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## Create Segment:

The screenshot displays the PyLab website with a dark theme. The top navigation bar includes links for Home, Lessons, Features, How It Works, Code Editor, Sign In, and a Get Started button. Below the navigation bar, there are three tabs: Learn, Practice, and Create, with the Create tab being the active one. The main content area is titled 'Create Your Own Projects' and includes the subtitle 'Build meaningful applications with your Python skills'. It features two columns of text: the left column encourages working on complete projects like web scrapers and games, while the right column states that building real projects is the best way to cement knowledge. A code editor window is open on the right, showing a Python script for a simple weather app project. The code includes imports for requests, a function to fetch weather data from an API, and a main block that calls the function for New York and prints the result.

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Learn Practice **Create**

### Create Your Own Projects

Build meaningful applications with your Python skills

Put everything together by working on complete projects like web scrapers, data analysis tools, games, and more. Save your work, share it with others, and build a portfolio.

Building real projects is the best way to cement your knowledge and demonstrate your skills to others.

```
weather_app.py
Simple weather app project
import requests

def get_weather(city):
 api_key = "YOUR_API_KEY"
 url = f"https://api.example.com/weather?city={city}&key={api_key}"
 response = requests.get(url)
 return response.json()

weather = get_weather("New York")
print(f"Current temperature: {weather['temp']}°C")
```





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**Thank You!**

